

# Fair Federated Learning on Real Hospital Data: FairTrade on FLamby Fed-Heart-Disease

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Code and per-seed results: [github.com/JordanManoj/fair-fl-medical](https://github.com/JordanManoj/fair-fl-medical) (results at commit 7080ef0)

**ABSTRACT.** Fair federated learning methods are mostly evaluated on census-style benchmarks with synthetic client splits. This report tests them on a real cross-silo medical federation: FLamby's Fed-Heart-Disease, with four hospitals, 740 patients and sex as the sensitive attribute. I reproduce FLamby's baselines exactly (all per-seed local-model accuracies match the published files), port FairTrade (AAAI 2024) to the per-hospital loaders, and run every comparison over 10 seeds with 95% confidence intervals and validation-only model selection. Four findings stand out. (1) FLamby's pooled-test convention overstates federated accuracy (0.748 vs 0.705 under the normalization clients actually use). (2) Every accuracy gain, from federated normalization to pooled training, widens the sex gap. (3) Because heart disease is far more common among men in this data (60% vs 25%), demographic parity over-diagnoses women; equal opportunity is the better target and covers significantly more of the trade-off space ( $p = 0.039$ ). (4) FairTrade's per-client fairness penalty switches off entirely when no client sees both sexes; computing it from four securely aggregatable group totals per client keeps it working (hypervolume +0.081,  $p < 0.001$ ), at no significant cost elsewhere.

## 1 Question and setting

Hospitals cannot pool patient records, so federated learning (FL) trains one model by averaging locally trained updates [3]. A shared model can still treat patient groups unequally, and FL adds a difficulty centralized training lacks: each hospital sees only its own patients, so it may see few or no members of a protected group. FairTrade [2] addresses fairness in FL with a fairness-penalized local objective and multi-objective Bayesian optimization (MOBO) that returns Pareto-optimal accuracy–fairness trade-offs. The question here is whether that machinery survives a real cross-silo medical federation, and what has to change where it does not.

**Data.** Fed-Heart-Disease [1, 5] has four hospitals with FLamby's fixed train/test split. The model input has 13 features (FLamby drops three UCI columns and one-hot encodes two). Sex is both an input and the sensitive attribute (female = protected). Table 1 shows why this data is a hard test.

| Hospital      | Train | Test | Women (train / test) | Disease rate | Cholesterol recorded as 0 |
|---------------|-------|------|----------------------|--------------|---------------------------|
| Cleveland     | 199   | 104  | 63 / 34              | 46%          | 0%                        |
| Hungary       | 172   | 89   | 50 / 19              | 38%          | 0%                        |
| Switzerland   | 30    | 16   | 3 / 0                | 100%         | 100%                      |
| Long Beach VA | 85    | 45   | 2 / 3                | 78%          | ≈25%                      |

**Table 1.** Fed-Heart-Disease per hospital (disease rate on the training split). Over all 740 patients, 24.7% of the 174 women and 60.1% of the 566 men have heart disease.

**Protocol.** Unless stated otherwise, results are on the union of the four test sets (254 patients, 56 women; 15 of them sick), each hospital's data normalized as it was in training. Reported metrics: balanced accuracy, statistical parity difference  $SPD = P(\hat{y}=1 \mid F) - P(\hat{y}=1 \mid M)$ , and equal opportunity difference  $EOD = TPR_F - TPR_M$  [4]. Undefined metrics (a group absent from a test set) are reported as n/a, never 0. Every configuration runs over 10 seeds (42–51) and reports mean  $\pm$  95% t-interval. Comparisons are paired by seed. All preprocessing statistics are fit on training data only. For tuned methods, each hospital's training split is further divided 75/25 into fit and validation parts, and *all* selection uses validation metrics only.

## 2 Baselines and two evaluation pitfalls

**Reproduction.** With FLamby's code and hyperparameters, FedAvg reaches  $0.748 \pm 0.027$  (std) on FLamby's pooled test, identical to the published value. All 100 published per-seed accuracies of the local models (4 models  $\times$  5 test sets  $\times$  5 seeds) are matched exactly, as are 24 of 25 FedAvg values (one differs by a single patient).

**Pitfall 1: pooled-test normalization.** FLamby normalizes its pooled test set with statistics of all training data combined, which no FL client has. Evaluated under the per-hospital normalization the model was trained with, the same FedAvg models score 0.705, not 0.748. The same mismatch affects FLamby's per-hospital numbers for the pooled-training baseline (e.g. Long Beach: 0.644 reported vs 0.822 with consistent normalization).

**Pitfall 2: accuracy at high-prevalence sites.** Sharing only per-hospital counts, sums and sums of squares lets all hospitals normalize with global statistics. This raises balanced accuracy by  $+0.060 \pm 0.019$  (10/10 seeds), and imputing the zero-coded missing cholesterol values rules out a "missing = sick" shortcut. A leave-one-hospital-out test shows the gain partly generalizes to unseen hospitals with balanced disease rates ( $+0.025$  Cleveland,  $+0.029$  Hungary, both  $p < 0.05$ ). At Long Beach (78% sick), however, accuracy rises by 11 points while balanced accuracy *falls* by 6 ( $p = 0.004$ ). Accuracy alone is therefore misleading here, and balanced accuracy is used throughout.

| Model (10 seeds, all-centers test)                  | Balanced accuracy | SPD (F – M)        | EOD (F – M)        |
|-----------------------------------------------------|-------------------|--------------------|--------------------|
| FedAvg, FLamby settings, per-hospital normalization | $0.704 \pm 0.009$ | $-0.160 \pm 0.096$ | $-0.049 \pm 0.121$ |
| FedAvg, FLamby settings, federated normalization    | $0.764 \pm 0.014$ | $-0.242 \pm 0.078$ | $-0.111 \pm 0.102$ |
| Pooled training (no privacy; upper bound)           | $0.795 \pm 0.002$ | $-0.322 \pm 0.003$ | $-0.156 \pm 0.003$ |
| Cleveland alone                                     | $0.736 \pm 0.006$ | $-0.199 \pm 0.019$ | $-0.119 \pm 0.043$ |

**Table 2.** The more accurate the model, the larger the sex gap: better use of the data amplifies the difference in disease rates already present in the labels.

## 3 FairTrade on real hospitals

**Port.** Each client minimizes  $BCE + \alpha \cdot \Sigma_g(m_g - m)^2$ , where  $m_g$  is group  $g$ 's mean predicted risk and  $m$  the overall mean. Training uses full-batch local Adam (30 rounds  $\times$  10 steps) with size-weighted FedAvg aggregation. MOBO (8 Sobol + 16 qLogNEHVI [6] steps) searches over ( $\alpha$ , learning rate) to maximize validation balanced accuracy and fairness. A unit test confirms that the ported penalty equals the original DemographicParityLoss numerically. The baseline "FedAvg" in this section is the same trainer with  $\alpha = 0$  and a validation-tuned learning rate. It reaches 0.787, higher than FLamby's FedAvg, so every comparison below uses the same trainer for both methods.

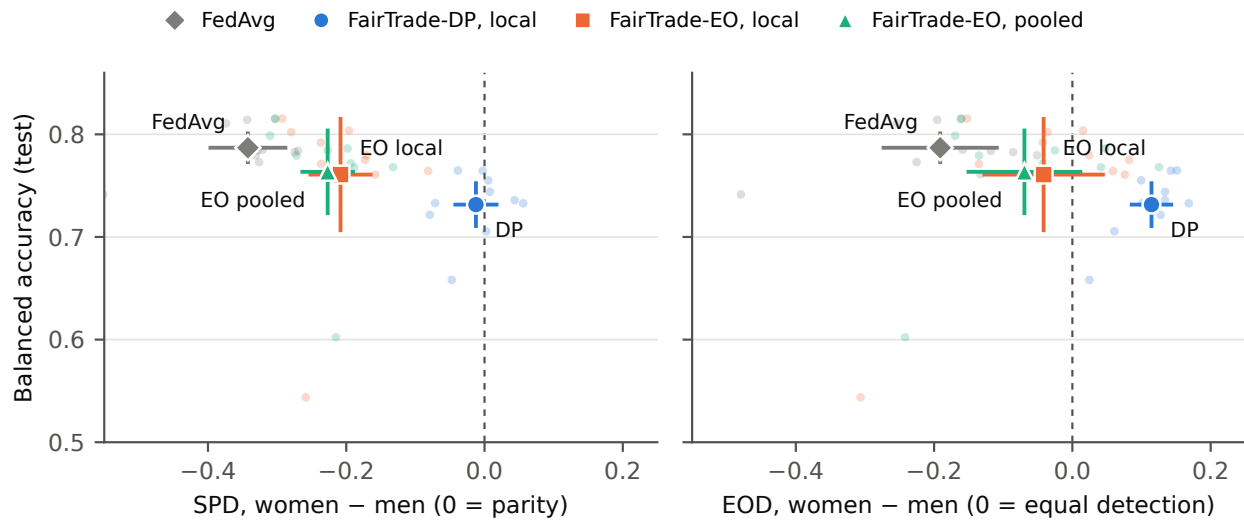
### IMPLEMENTATION ISSUES FIXED IN THE PORT

My challenge implementation of FairTrade had six problems, all fixed here. (i) MOBO objectives were computed on the test set. (ii) The hypervolume reference point lay above every attainable objective, so every hypervolume improvement was zero and the search was effectively random. (iii) Every candidate evaluation also updated one shared global model. (iv) Probabilities passed through two sigmoids. (v) Clients were averaged without size weights. (vi) The ATE option referenced an undefined class.

**Result.** FairTrade removes the sex gap in prediction rates: SPD goes from  $-0.342$  to  $-0.012 \pm 0.033$ , at a cost of 5.5 balanced-accuracy points ( $0.787 \rightarrow 0.732$ ). Its Pareto front nearly triples the test hypervolume in the (balanced accuracy,  $-|SPD|$ ) plane ( $0.047 \rightarrow 0.132$ ). But it pushes EOD to  $+0.114 \pm 0.031$ : sick women are now detected *more* often than sick men. This is the expected incompatibility of parity and error-rate fairness when base rates differ [4]. With disease rates of 25% and 60%, equal prediction rates can only be reached by over-diagnosing women.

## 4 Equal opportunity and pooled group totals

I varied FairTrade along two axes. **Notion:** demographic parity (DP) or equal opportunity (EO), the same penalty restricted to truly sick patients. **Scope:** *local* (FairTrade: each client measures fairness on its own patients) or *pooled*. In the pooled scope, at the start of each round every client reports per-group ( $\Sigma$  predicted risk, count) under the current global model, four numbers that secure aggregation can protect. Each client then penalizes the gap between the *global* group means, differentiating its own part and holding the other clients' part fixed. A unit test checks that this equals the penalty on all data combined. Related ideas, deriving fairness from aggregated group statistics, appear in prior fair-FL work such as FairFed [7]. The contribution here is evaluating the idea inside FairTrade on real hospital data. With only about 6 sick women per validation split, thresholded EOD moves about 16 points per patient. Model selection therefore uses soft gaps (differences in mean predicted risk), while test results are reported with the usual thresholded SPD and EOD.



**Figure 1.** Operating points (most accurate candidate with validation  $|\text{soft gap}| \leq 0.10$ ) on the all-centers test set; large markers with 95% CIs, small dots = individual seeds. DP reaches SPD  $\approx 0$  but overshoots to EOD  $\approx +0.11$ . EO closes the detection gap while keeping more accuracy. Its remaining SPD of about  $-0.21$  reflects the real difference in disease rates. Numbers in Table 3.

| Variant (10 seeds)    | Balanced accuracy | SPD      | EOD      | HV (bacc, $- \text{SPD} $ ) | HV (bacc, $- \text{EOD} $ ) |
|-----------------------|-------------------|----------|----------|-----------------------------|-----------------------------|
| FedAvg (same trainer) | $0.787 \pm 0.016$ | $-0.342$ | $-0.191$ | $0.047$                     | $0.090$                     |
| FairTrade-DP, local   | $0.732 \pm 0.023$ | $-0.012$ | $+0.114$ | $0.132$                     | $0.127$                     |
| FairTrade-DP, pooled  | $0.747 \pm 0.010$ | $-0.013$ | $+0.114$ | $0.132$                     | $0.133$                     |
| FairTrade-EO, local   | $0.761 \pm 0.056$ | $-0.208$ | $-0.042$ | $0.101$                     | <b>0.137</b>                |
| FairTrade-EO, pooled  | $0.763 \pm 0.042$ | $-0.227$ | $-0.069$ | $0.103$                     | $0.131$                     |

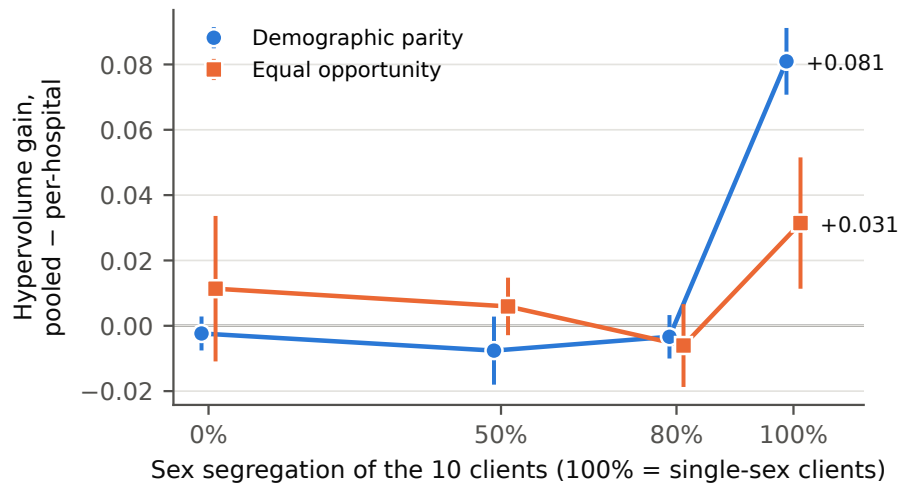
**Table 3.** HV = test hypervolume of the validation Pareto front, reference point  $(0.5, -0.5)$ . Paired over seeds: EO vs DP (local) in the EOD space  $+0.011 \pm 0.010$  ( $p = 0.039$ , 9/10 seeds); pooled vs local: not significant for either notion ( $p \geq 0.10$ ).

Equal opportunity is the better target for this diagnostic task. On the real hospitals, however, pooled totals bring no significant gain. 96% of the women in the fit data are at Cleveland and Hungary (on average 48 and 38 per seed, vs 2.5 at Switzerland and 1.2 at Long Beach), so those two hospitals' local penalties already carry almost all of the fairness signal.

## 5 Stress test: when no client sees both sexes

To test the failure mode the pooled penalty targets, I re-split the same fit patients into 10 clients. With probability  $s$  a patient goes to a sex-specific client (women to clients 0–1, men to 2–9), otherwise to a random one. Validation and test sets remain the original hospital splits. Every variant gets the same  $\alpha$  grid

{1, ..., 1000} at FedAvg's validation-selected learning rate, which isolates the penalty scope from the tuner. This partition is motivated by real single-sex sites such as veterans' hospitals and women's clinics. Long Beach VA already has a single woman in its fit split.



**Figure 2.** Test hypervolume gain of the pooled over the local penalty (own notion's space), mean and 95% CI over 10 seeds. Clients with both sexes: 10, 9.9, 8.6 and 0 of 10 at  $s = 0, 0.5, 0.8, 1$ . Gains at  $s = 1$ : DP  $+0.081 \pm 0.010$  ( $p < 0.001$ , 10/10 seeds), EO  $+0.031 \pm 0.020$  ( $p = 0.006$ , 8/10); all gains at  $s \leq 0.8$  are not significant ( $p \geq 0.13$ ).

At  $s = 1$  the local penalty is identically zero, and FairTrade's results equal FedAvg's *exactly*. The pooled penalty still works: under EO it reduces the detection gap from  $-0.213$  to  $-0.061$  at nearly unchanged balanced accuracy ( $0.790 \rightarrow 0.785$ ). The failure is a cliff, not a slope. At  $s = 0.8$  the two women-heavy clients still hold about 10 men between them, which is enough for local penalties to steer the federation. Across all eight comparisons, the pooled penalty is never significantly worse, which makes it a safe default.

## 6 Limitations

- **Small groups.** The test set has 15 sick women, so test EOD moves about 7 points per patient, and the CIs reflect training variability, not test-sampling noise. In one seed, EO selection picked  $\alpha$  at the search bound (balanced accuracy 0.54–0.60).
- **Semi-synthetic stress test.** The segregated partitions use real patients but synthetic client assignment. A naturally segregated federation is needed to confirm it.
- **Scope.** One small tabular dataset (13 features, logistic regression), one binary sensitive attribute, honest-but-curious privacy assumptions for the four shared numbers, and no formal differential-privacy accounting.
- **Novelty.** I have not yet done a systematic literature review of aggregated-statistics fairness in FL.

## 7 Proposed next steps (thesis directions)

1. **A naturally segregated federation:** eICU (200+ US ICUs with sex, age and ethnicity) to test whether the pooled penalty matters at realistic scale and heterogeneity.
2. **Robust selection under tiny groups:** cross-validated or Bayesian estimates of EOD for MOBO, plus test-set bootstrap intervals.
3. **Privacy of the shared totals:** secure aggregation and differentially private group counts, and their effect on the trade-off.
4. **Imaging:** Fed-ISIC2019 (skin lesions, 6 sources) to move beyond logistic regression.

## REPRODUCIBILITY

Python 3.11, pinned versions in `requirements.txt`, FLamby pinned at `edacf54`. One command sets up the environment and data (`scripts/setup.sh`). Each table above has its own script (`run_fedavg.py`, `run_baselines.py`, `run_loho.py`, `run_fairtrade.py`, `compare_fairtrade.py`, `run_stress.py`) and per-seed CSVs in `results/`; the figures are generated by `make_figures.py`. A full run log with every decision is in `results/LOG.md`, and 10 unit tests run with `pytest`.

## References

1. J. Ogier du Terrail et al. FLamby: Datasets and benchmarks for cross-silo federated learning in realistic healthcare settings. *NeurIPS Datasets and Benchmarks*, 2022.
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